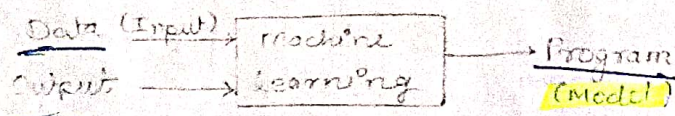
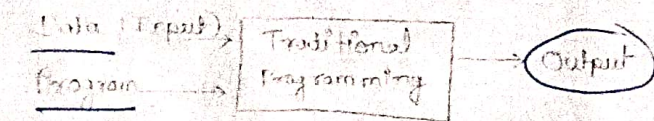


ML

Machine Learning

* Capability to learn without being explicitly programmed
(without being actually programmed)



* Data :- fact, figs, statistics, observations.

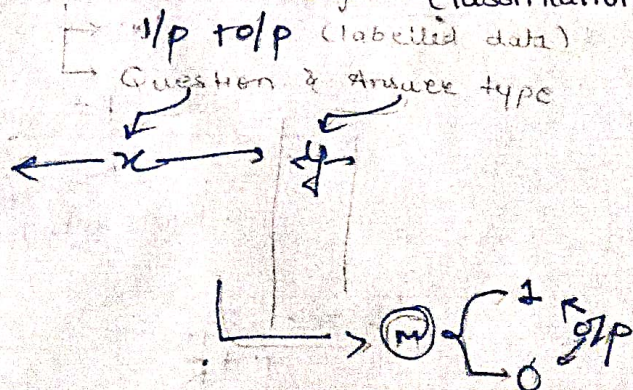
* Information :- meaningful, useful, relevant data

* Labelled data: $\frac{I/p}{Q} + \frac{o/p}{A} \rightarrow$ supervised learning

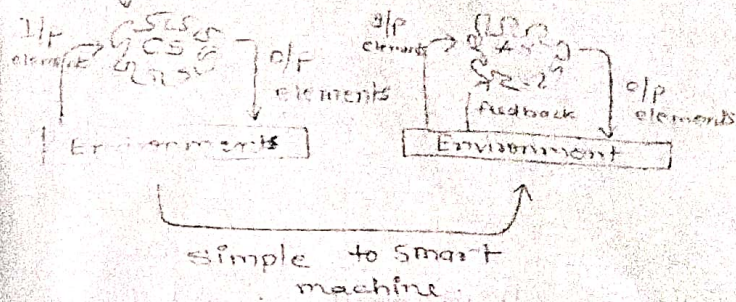
* Types of Learning:

- ① Supervised Learning
- ② Unsupervised Learning
- ③ Reinforcement Learning

① Supervised Learning: $\begin{cases} \text{Regression} \\ \text{Classification} \end{cases}$



* Classic & Adaptive machines



* Machine Learning Cycle:

- ① Problem Understanding
- ② Data collection
- ③ Data preparation
- ④ Model selection
- ⑤ Model building
- ⑥ Model Evaluation
- ⑦ Model Tuning
- ⑧ Deployment
- ⑨ Monitoring

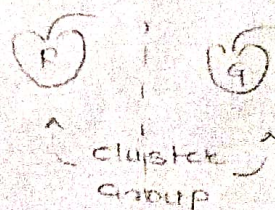
② Unsupervised Learning:

> Only I/p no output

> only Question type

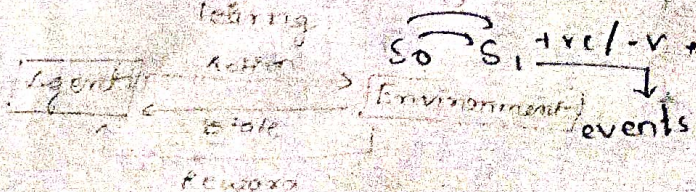
> Unlabelled data

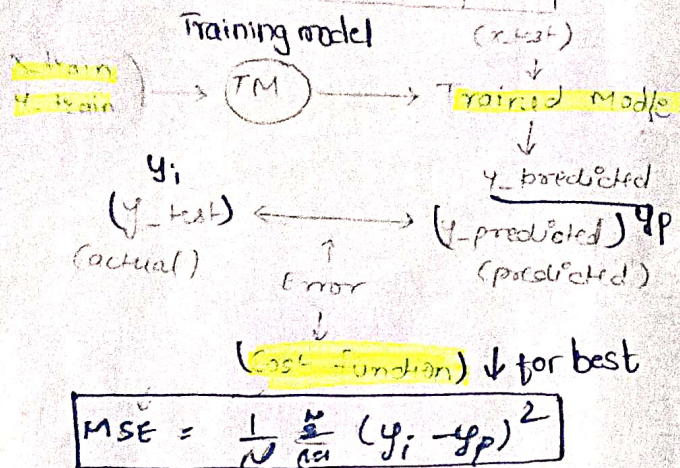
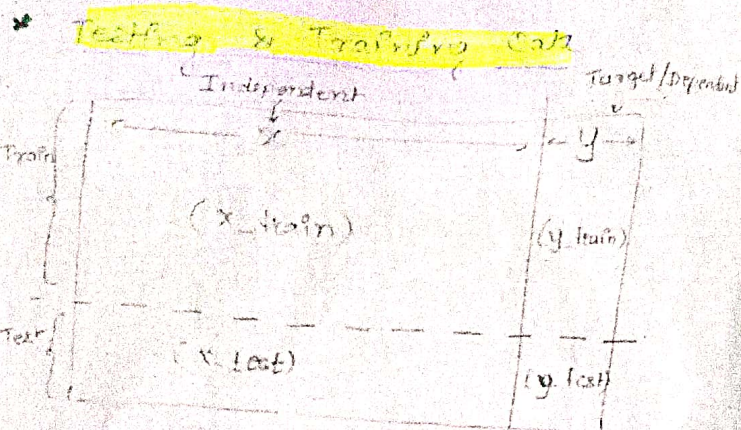
> Look out for similarity



③ Reinforcement Learning

> Rewards (penalty) based learning





Confusion Matrix

		Predicted		
		NO	Yes	
Actual	NO	50 [TN]	10 [FP]	60
	Yes	5 [FN]	100 [TP]	105
		55	110	

TN ↑ TP ↑
↓
for better accuracy

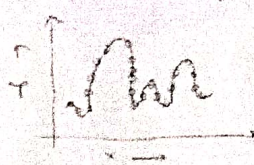
↑ **Accuracy** = $\frac{TP + TN}{TN + TP + FN + FP} = \frac{100 + 50}{165} = 0.81 \checkmark$

↓ **Error** : 1 - Accuracy
or
 $\frac{FP + FN}{Total} = 0.09 \checkmark$

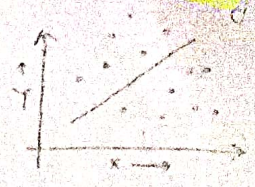
↑ **Precision** : $\frac{TP}{TP + FP} = \frac{100}{110} = 0.909 \text{ (1)}$
↳ it is for predicted

↑ **Recall** : $\frac{TP}{Actual\ Yes} = \frac{TP}{TP + FN} = \frac{100}{105} = 0.95 \text{ (1)}$
↳ it is for actual

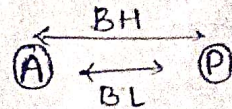
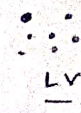
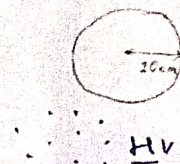
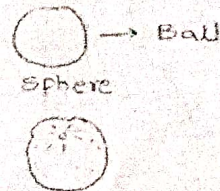
Overfitting



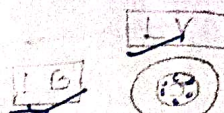
Underfitting



- Sphere
- play
- Eat
- Radius = 5cm



Bias & Variance



← Ideal



Training test & overfitting

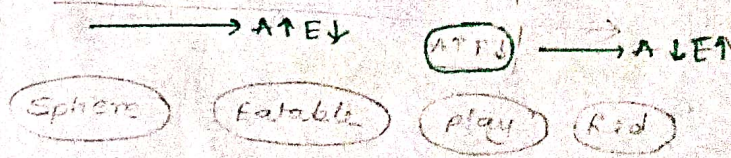
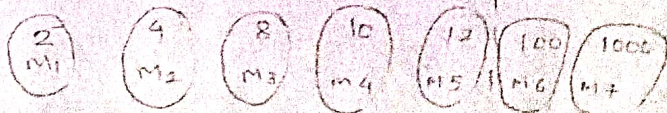


underfitting



Not too many

Curse of Dimensionality



Precision & Recall Harmonic mean

F1 Score

TP = 50
FP = 10
TN = 10
FN = 30
P = 0.83
R = 0.62

	TP	FP
	FN	TN

$P = \frac{TP}{TP + FP}$
 $R = \frac{TP}{TP + FN}$

$F1 = \frac{2 \cdot P \cdot R}{P + R}$
total number

$F1 = \frac{2}{\frac{1}{P} + \frac{1}{R}}$

TN  FP
 $TN + FP$ TN
 ↳ for actual negative
 $Recall = \frac{TP}{TP + FN}$

↳ for actual negative

Recall of TP
+ Pto

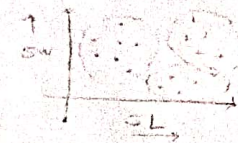
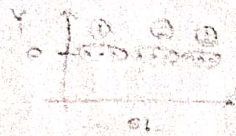
x Uni 31 Mull

Variate Analysis






- selosa
- versicolor
- virginica



SL L L L L
SW L L L L
PL L L L L
PW L L L L

④ Label Encoding (categorical value in order)

$$g = \frac{5}{2}$$

Pune	→	0
Goa	→	1
Mumbai	→	2
Pune	→	0
Punjab	→	3
Goa	→	2

Pune $\rightarrow 0$
Goa $\rightarrow 1$
Mumbai $\rightarrow 2$
Punjab $\rightarrow 3$

(low wgh medru)
↓ ↓ ↓
0 1 2

⑤ One hot encoding (not words)



Pine	1	0	0	0
Gee	0	1	0	0
Mombas	0	0	1	0
Pure	1	0	0	0
Pumpkin	0	0	0	1
Gee	0	1	0	0

Prevalence

① Normalization (Minimum Error)

$\chi = \chi - \chi_{\text{proy}}$
 $\chi_{\text{proy}} = \chi_{\text{proy}}$

② standardization [standard scores]

$(\mu \rightarrow 0 \text{ \& } \sigma \rightarrow 1)$

$$x' = \frac{x - u}{\sigma}$$

③ Robust Scale

TQR (Tetraquidile Range)

$\Rightarrow 10, 20, 30, 40, 45, 50, 55, 60, 70, 80$

$$1^{st} \text{ Quotient} = 25\% \quad P = 30$$

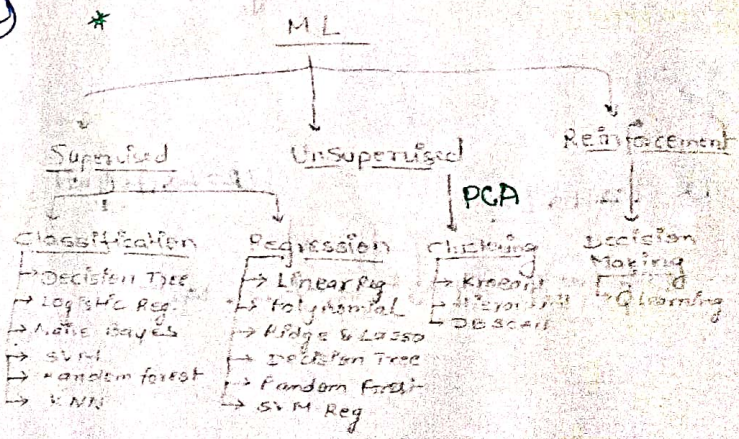
2nd Quartile = 50% P = 47.5

3rd Quartile = 75% P = 58.75

4th Quartile = 100% F = 20

$$x' = \frac{x - x_{mid}}{Q_b - Q_1}$$

✱



* Regression

→ Continuous data

↳ Relationship b/w Independent & dependent variable ✓

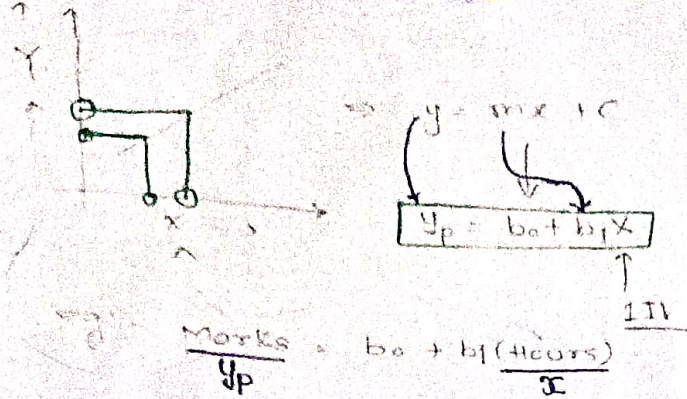
⑦ StandardScaler (Standard Scaler)

✓ Green 11 11

✓ 1919

multicollinearity: Dependency of independent variable metrics

Simple Linear Regression



Multiple Linear Regression

x_1	x_2	x_3	y
Hours	Attend	Assign	Marks

Marks = $b_0 + b_1(\text{Hours}) + b_2(\text{Attendance}) + b_3(\text{Assignment})$

helps in Feature selection

Lasso Regression

Ridge Regression

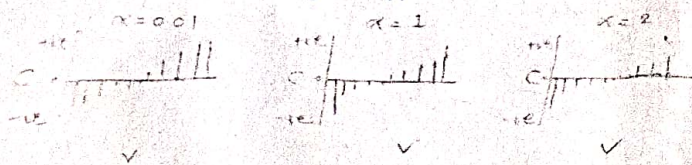
Regularization

RR $\text{Loss} + \alpha \|w\|^2$

(penalty)

coefficient

$\|w\|^2 = w_1^2 + w_2^2 + w_3^2 + \dots + w_n^2$



$y = 0.4 + 1.2x_1 + 2.0x_2 + 2.9x_3$

$y = 0.4 + 0.2x_1 + 2.0x_2 + 5.0x_3$

here the value is reduced but the impact is same

Elastic Net Regression (Hybrid)

RR $\text{Loss} + \alpha \|w\|^2$

LR $\text{Loss} + \alpha \|w\|$

EN $\text{Loss} + \alpha_1 \|w\|^2 + \alpha_2 \|w\|$

12 ratio: how to mix L and LR
0 + large
0.5 + balance

(multicollinearity) RR

(feature selection) LR

alpha controls regularization strength
alpha up -> more penalty (helps to reduce overfitting)

Polynomial Regression



MAE -> ↓
MSE -> ↓
RMSE -> ↓
R^2 -> ↑
for Better model

$y = b_0 + b_1 x$

$y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4$

$y = \alpha_0 + \sum_{i=1}^n b_i x_i^d$

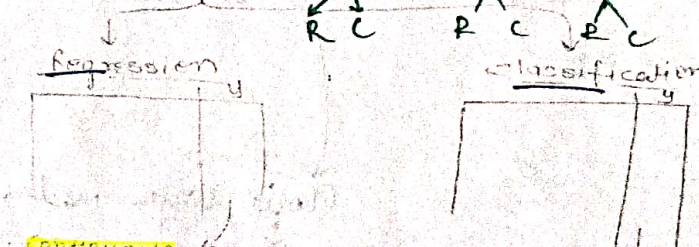
- ✓ degree poly: $y = \text{constant}$
- ✓ degree poly: $y = mx + c$
- ✓ degree poly: $y = ax^2 + bx + c$

$y = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \dots + b_n x^n$

$y = b_0 + \sum_{i=1}^n b_i x_i + \text{FP}$

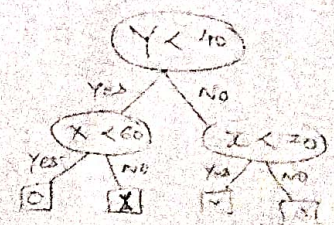
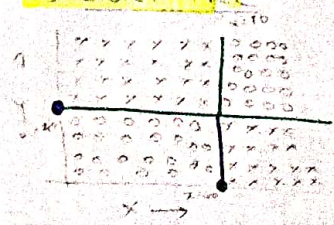
polynomial features

Decision Tree, SVM, KNN, Random forest



continuous (Mean/Average)
categorical (Majority)

Decision Tree



Entropy (E) = $-\text{Pr}(\log_2 \text{Pr}) - \text{Pr}(\log_2 \text{Pr})$

calculated for a particular node
probability entropy

probability of class
probability of class

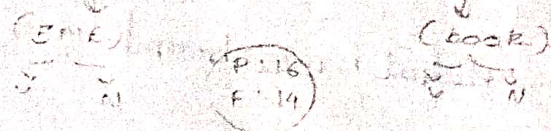
551
200

505

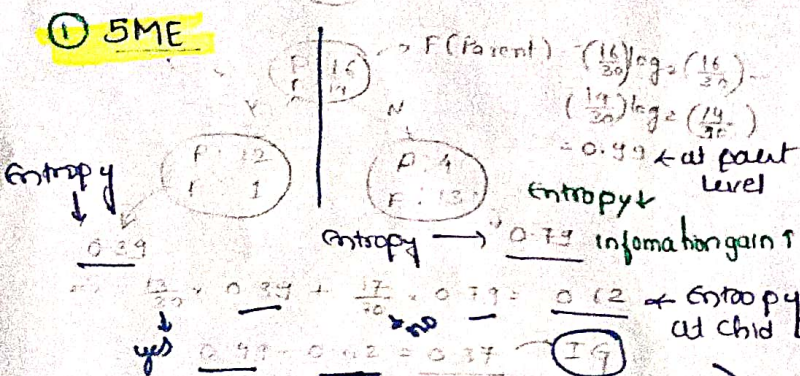
20)

$\Rightarrow 0.85$

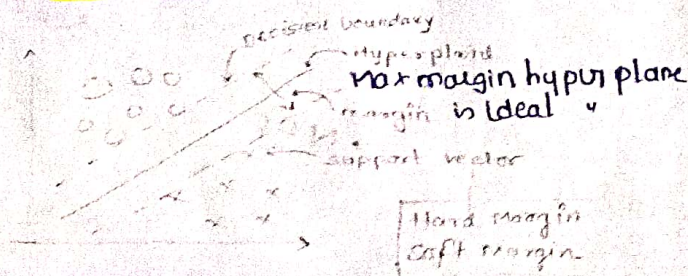
eg:- 2 features



① 5ME



*CVM: Support Vector Machine

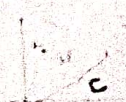


→ Binary classification

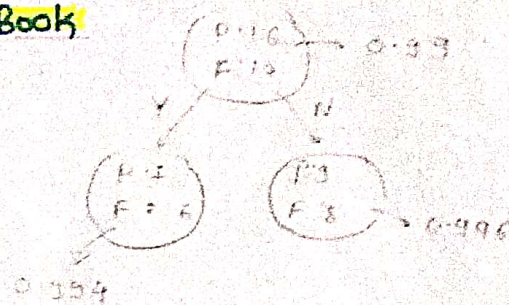
Hyperplane $y^* = w^T x + b = 0$

normal vector to hyperplane
(direction $\perp$)

1. Prüfung $\frac{1}{2}$ $\rightarrow$ $w^T x + b \geq 0$
 2. $\text{Sukz.$ $\frac{1}{2}$ $\rightarrow$ $w^T x + b < 0$


$$\begin{aligned} \gamma &= \frac{1}{2} \left(\frac{1}{\alpha} + \alpha \right) \\ \gamma &= \frac{1}{2} \left(\frac{1}{\alpha} + \alpha \right) \\ \gamma &= \frac{1}{2} \left(\frac{1}{\alpha} + \alpha \right) \end{aligned}$$

② Book



$$= \frac{12}{20} \times 0.44 + \frac{17}{20} \times 0.496$$

10

0 0 7

0 0 7

[0 0 7]

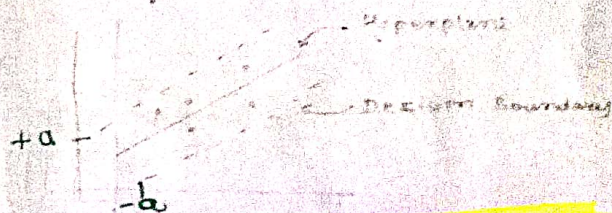
→ $\frac{1}{2} \times 100 = 50\%$

(NO scaling)



where $d_1, d_2, \dots, d_n$ are data sample
and $m_1, m_2, \dots, m_n$ are mode found with it

*SVR (Regression)



best upper bound that has max number of points.

47-100

$$C_1 Y = Wx + b$$

7-1-DB

$$\begin{aligned} Wx + b &= +a \\ Wx + b &= -a \end{aligned}$$

Range $\rightarrow -a < Y_{wxtb} < +a$

(very good)

Linear regression

Linear regression

Linear regression

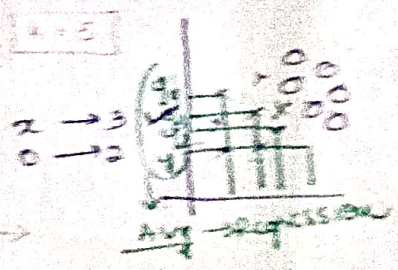
Linear regression

Linear regression



$$K(x, x') = x x' [f(w) w]$$

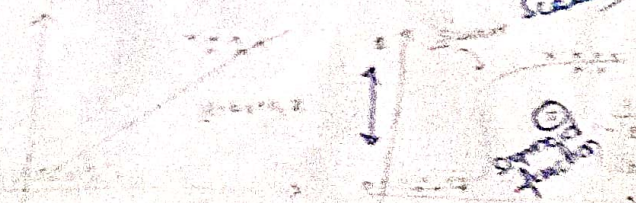
$$K(x, x') = (1 + x x')^p \quad (\text{min max})$$



→ maximization (maximizing)
→ regression (average/mean)

1 1 1 1 1
1 1 1 1 1

static regression (standard scales)



input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

input features

$$P(\text{Car}) = \frac{2}{10} = 0.2$$

$$P(\text{Bike}) = \frac{7}{10} = 0.7$$

$$P(\text{Car} | J, B) = \frac{P(J|\text{Car}) \cdot P(B|\text{Car}) \cdot P(\text{Car})}{P(J) \cdot P(B)}$$

$$P(J) \cdot P(B)$$

$$\frac{2}{10} \times \frac{2}{10} \times \frac{3}{10}$$

$$\frac{2}{10} \times \frac{2}{10} = 0.066$$

$$P(\text{Bike} | J, B) = \frac{P(J|\text{Bike}) \cdot P(B|\text{Bike}) \cdot P(\text{Bike})}{P(J) \cdot P(B)}$$

$$P(J) \cdot P(B)$$

$$\frac{2}{7} \times \frac{5}{7} \times \frac{7}{10}$$

$$= 0.142$$

★ Gaussian NB

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\text{Yes} \rightarrow Y_{\text{Yes}} [80, 70, 90, 75, 70, 86, 96, 80, 69]$$

$$\text{No} \rightarrow Y_{\text{No}} [40, 30, 51, 35, 25]$$

$$\mu(Y) \rightarrow 73.1$$

$$\sigma(Y) \rightarrow 10.2$$

$$\mu(N) \rightarrow 26.2$$

$$\sigma(N) \rightarrow 3.7$$

ED for (3) :-

$$k1 \rightarrow \sqrt{(168-125)^2 + (60-72)^2}$$

$$d1 \rightarrow 20.80$$

$$k2 \rightarrow \sqrt{(168-170)^2 + (60-56)^2}$$

$$d2 \rightarrow 4.42$$

$$k \quad d2 < d1$$

③ → $k2$ ✓ *needed to find centroid for the next iteration*

new centroid calculation:

$$\text{for } k2 = \left(\frac{170+168}{2}, \frac{60+56}{2} \right) = (169, 58)$$



$$d1 \rightarrow \sqrt{(174-125)^2 + (68-72)^2}$$

$$d1 \rightarrow 7.21$$

$$d2 \rightarrow \sqrt{(174-169)^2 + (68-58)^2}$$

$$d2 \rightarrow 10.19$$

$$\text{Ans: } k1 \rightarrow \{1, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$k2 \rightarrow \{2, 3\}$$

$$P(\text{Marks} = 74 | \text{SMC} = Y) \downarrow$$

$$= \frac{1}{\sqrt{2\pi}(10.2)} e^{-\frac{(74-73.1)^2}{2(10.2)^2}}$$

$$= 0.034$$

$$P(\text{Marks} = 74 | \text{SMC} = N) \downarrow$$

$$= \frac{1}{\sqrt{2\pi}(9.7)} e^{-\frac{(74-26.2)^2}{2(9.7)^2}}$$

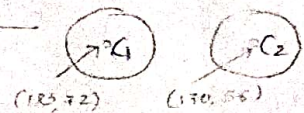
$$= 0.018$$

★ Unsupervised Learning

① K-means

$$d, cd, d1 \quad \frac{1}{\text{similarity}}$$

	Height	Weight
①	185	72
②	170	56
③	168	60
④	173	62
⑤	122	72
⑥	182	77
⑦	180	71
⑧	180	70
⑨	183	84
⑩	180	82
⑪	180	87
⑫	177	76



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$n_1 \xrightarrow{d1} c_1$$

$$n_2 \xrightarrow{d2} c_2$$

② K-medoids clustering K=2

(consider medoids)

m medoid

	x	y	k1	k2	cost	k1	k2	cost
①	2	2	0	0	0	0	0	0
②	3	5	2	2	2	8	6	6
③	4	7	4	3	3	9	7	7
④	2	4	2	4	2	2	0	2
⑤	5	5	6	0	0	6	4	4
					7			21

Manhattan distance

$$\Rightarrow |x_2 - x_1| + |y_2 - y_1|$$

$$= |5 - 2| + |5 - 2|$$

$$= |3 + 3| = 6$$

$$\Rightarrow |3 - 5| + |5 - 2|$$

$$= 5 + 3 = 8$$

$$\Rightarrow |3 - 5| + |5 - 5|$$

$$= 2$$

$$\Rightarrow |4 - 2| + |7 - 2|$$

$$= 2 + 5 = 7$$

$$\Rightarrow |4 - 5| + |7 - 5|$$

$$= 1 + 2 = 3$$

$$\Rightarrow |2 - 5| + |4 - 2| = 2$$

$$= |3 - 5| + |4 - 5| = 4$$

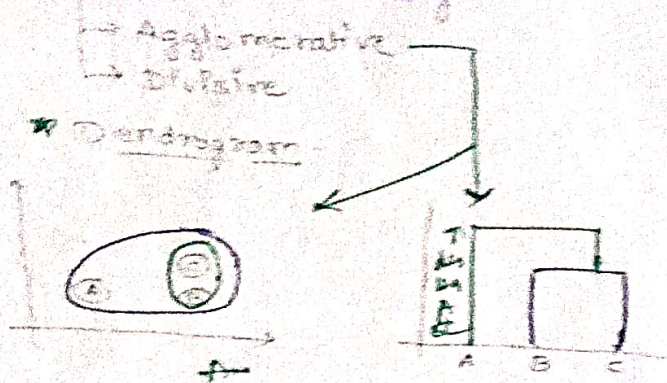
cost → minimum distance



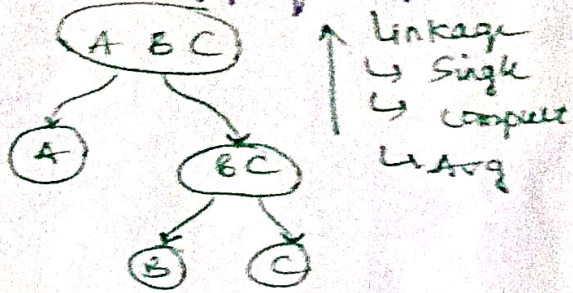
chasing the medoid

Unit 10

Agglomerative & Divisive



Divisive is opp of Agglomerative



Linkage
→ Single
→ Complete
→ Avg

Ex. 10.1

X	Y
2.5	2.5
1.5	1.5
1.2	2.2
1.3	2.2
0.2	0.2
2.2	2.2
1	1.5
1.5	1.2
2.5	1.1
1.2	0.2

$\bar{X} = 2, \bar{Y} = 1.81$

consider below

$$Cov(X, Y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$C = \begin{bmatrix} Cov(X, X) & Cov(X, Y) \\ Cov(Y, X) & Cov(Y, Y) \end{bmatrix}$$

Nein cov
cov with pop

X	X - $\bar{X}$	(X - $\bar{X}$)(Y - $\bar{Y}$)
2.5	0.5	0.475
1.5	-0.5	-0.625
1.2	-0.8	-1.22
1.3	-0.7	-1.22
0.2	-1.8	-3.22
2.2	0.2	0.22
1	-1	-1.22
1.5	-0.5	-0.92
2.5	0.5	0.22
1.2	-0.8	-1.22

Sum = 5.5490

Y	Y - $\bar{Y}$	(Y - $\bar{Y}$)(X - $\bar{X}$)
2.5	0.69	0.2401
1.5	-0.31	-0.2401
1.2	-1.01	-0.8081
1.3	-0.91	-0.8081
0.2	-1.61	-2.5921
2.2	0.39	0.1521
1	-0.81	-0.6561
1.5	-0.31	-0.2401
2.5	0.69	0.2401
1.2	-1.01	-0.8081

Sum = 6.449

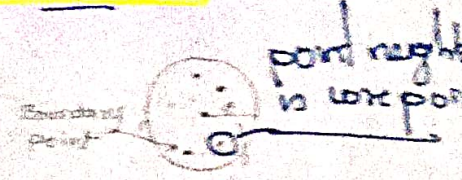
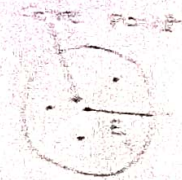
X	Y	X - $\bar{X}$	Y - $\bar{Y}$	(X - $\bar{X}$)(Y - $\bar{Y}$)
2.5	2.5	0.5	0.69	0.345
1.5	1.5	-0.5	-0.31	-0.155
1.2	2.2	-0.8	0.39	-0.312
1.3	2.2	-0.7	0.39	-0.273
0.2	0.2	-1.8	-1.61	2.898
2.2	2.2	0.2	0.39	0.078
1	1.5	-1	-0.81	0.81
1.5	1.2	-0.5	-0.31	0.155
2.5	1.1	0.5	-0.69	-0.345
1.2	0.2	-0.8	-1.01	0.808

Sum = 5.5490

Distance (for cluster data)

Distance is the distance between two points in a space.

Distance is the distance between two points in a space.



standard scaling

Principal Component Analysis

Dimension Reduction Techniques

Dependent variable

Dependent variable

Dependent variable

Dependent variable

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$$\begin{bmatrix} 0.7351 & 0.2649 \\ 0.2649 & 0.7351 \end{bmatrix} \Rightarrow \lambda_1 = 1.2890, \lambda_2 = 0.9900$$

$$\begin{bmatrix} 0.6798 & 0.7351 \end{bmatrix} \Rightarrow \lambda_1 = 1.2890, \lambda_2 = 0.9900$$

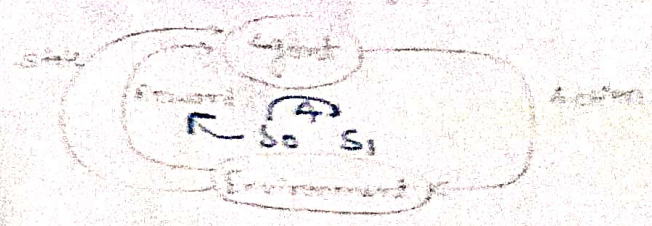
$$\text{Sum of } \lambda_1 \text{ \& } \lambda_2 = 1.2890 + 0.9900 = 2.2790$$

λ_1, λ_2 are the principal components

Value of state is total amount of reward an agent can expect to gain over the future starting from that particular state

- Initialize Q-table
- Choose an action
- Perform action
- Observe reward
- Update Q-table

Reinforcement Learning



Elements of RL

- Policy
- Value function
- Q-function

Policy defines agent's behavior for a given state

Value function provides a numerical value based on state of environment

Initialize Q-table

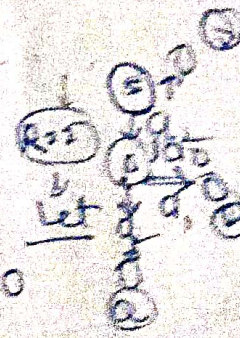
It is a simple lookup table where we calculate the expected future rewards for action at each state

Actions	a_1	a_2	a_3	a_4
State 1	0	0	0	0
State 2	0	0	0	0
State 3	0	0	0	0
State 4	0	0	0	0

Choose an action

3	0	0
0	0	0

Reward → Reaching goal = 10
Hole = -5
Plant = 0



* In above table reason
only the Max Q value
now all are zero

Update Q table

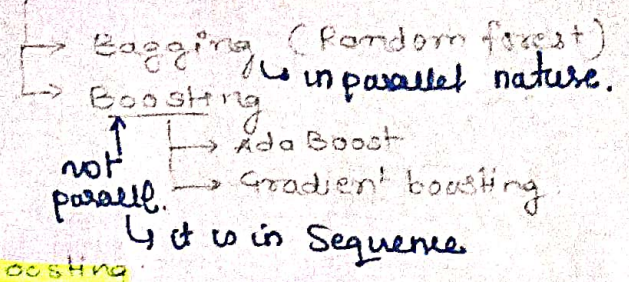
Using Q function

Leaves Bellman eqⁿ

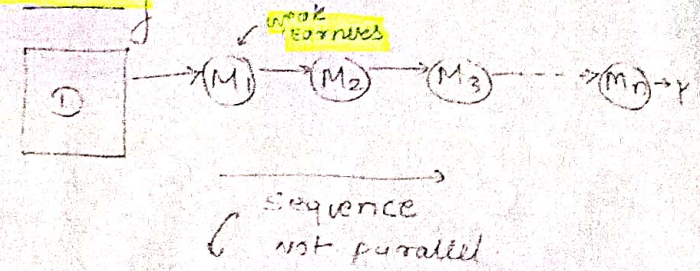
$$NQ(s,a) = Q(s,a) + \alpha [R(s,a) + \gamma \max_{a'} Q(s,a') - Q(s,a)]$$

new value ← $Q(s,a)$
 current value ← $Q(s,a)$
 Learning Rate ← α
 Reward for taking 'a' at 's' ← $R(s,a)$
 Discount Rate ← γ
 Max future reward ← $\max_{a'} Q(s,a')$

Ensemble Learning



Boosting



$$\alpha_1 M_1 + \alpha_2 M_2 + \alpha_3 M_3 + \dots + \alpha_n M_n$$

$\alpha \rightarrow$ weights / Importance / Influence

Ada Boost

① Dataset

①	→ 1/5
②	→ 1/5
③	→ 1/5
④	→ 1/5
⑤	→ 1/5

equal sample weight

② Get the best model

→ DT (highest 79)

③ Calculate the α

$$\alpha = \frac{1}{2} \log \frac{1 - TE}{TE} \quad \text{Miss-Classified}$$

$$\text{Performance} = \frac{1}{2} \log \frac{1 - TE}{TE}$$

$$\alpha = \frac{1}{2} \log \frac{1 - \frac{1}{5}}{\frac{1}{5}}$$

$$\alpha = 0.69$$

④ New SW

	SW
①	1/5
②	1/5
③	1/5
④	1/5
⑤	1/5

	NNSW
①	0.126
②	0.126
③	0.493
④	0.126
⑤	0.126

predicted wrong & weight

Take up to 1

$$NSW = \alpha W + e^{-\alpha}$$

'- ' → correct
'+' → wrong

$$NSW (\text{correct}) = \frac{1}{5} * e^{-0.69} = 0.1004$$

$$NSW (\text{wrong}) = \frac{1}{5} * e^{+0.69} = 0.3988$$

⑤ Error / bucket

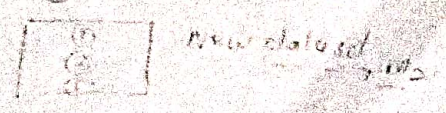
	NNSW	Error
①	0.126	0 - 0.126
②	0.126	0.126 - 0.252
③	0.493	0.252 - 0.745
④	0.126	0.745 - 0.871
⑤	0.126	0.871 - 0.997

weight

⑥ Random selection (0-1)

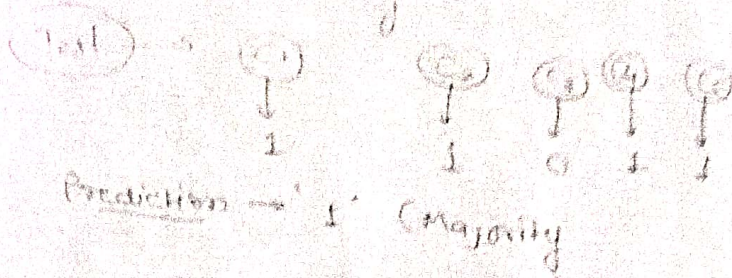
- 0.011 → ①
 - 0.33 → ②
 - 0.45 → ③
 - 0.60 → ④
 - 0.90 → ⑤
- 3 times

belongs max time

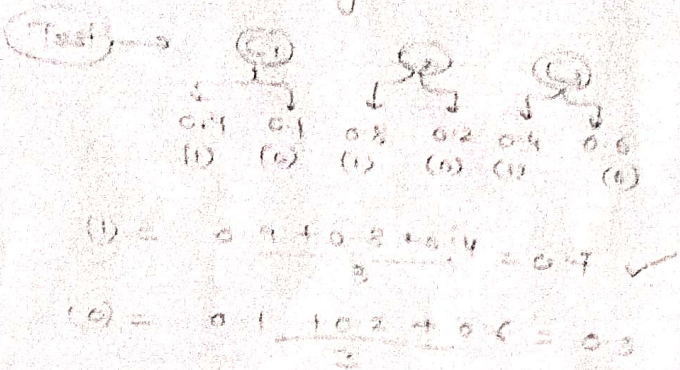


Majority Classification

↳ Hard Voting

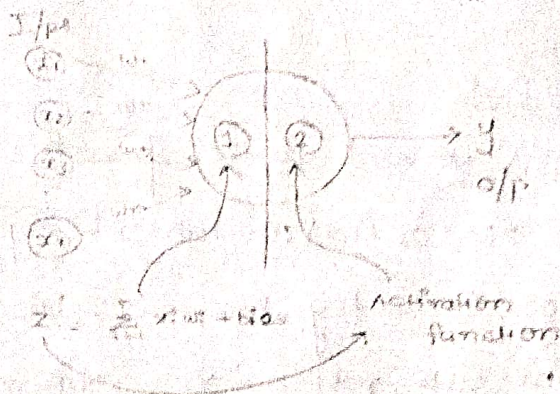


→ Soft Voting



Neural Network

↳ inspired human brain



Layers

- Input
- hidden
- output



Standard Deviation

Age	Income	P1	P2	P3	P4
25	15K	0.7	0.2	0.1	0.0
35	35K	0.4	0.5	0.1	0.0
45	45K	0.2	0.3	0.4	0.1
55	55K	0.1	0.2	0.3	0.4
65	65K	0.0	0.1	0.2	0.3

Mean

(1-4)

C. Each model produces a score

E. Takes the avg. (avg. of all)

I/p → (0.7, 0.4, 0.2, 0.1, 0.0)

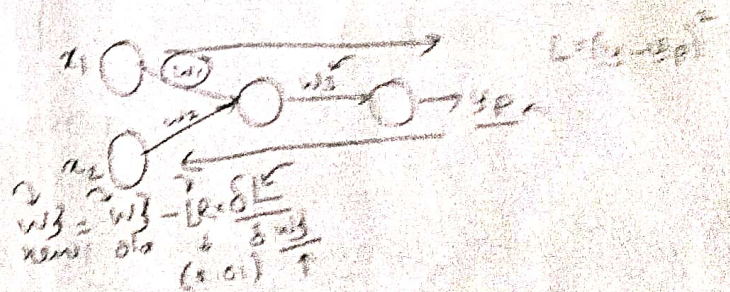
For I/p (0.7, 0.4, 0.2, 0.1, 0.0)

(2) Standard Deviation

σ = $\sqrt{\frac{1}{n} \sum (x_i - \mu)^2}$

For I/p (0.7, 0.4, 0.2, 0.1, 0.0)

Age	Income	P1
25	15K	0.7
35	35K	0.4
45	45K	0.2
55	55K	0.1
65	65K	0.0



Types of NN

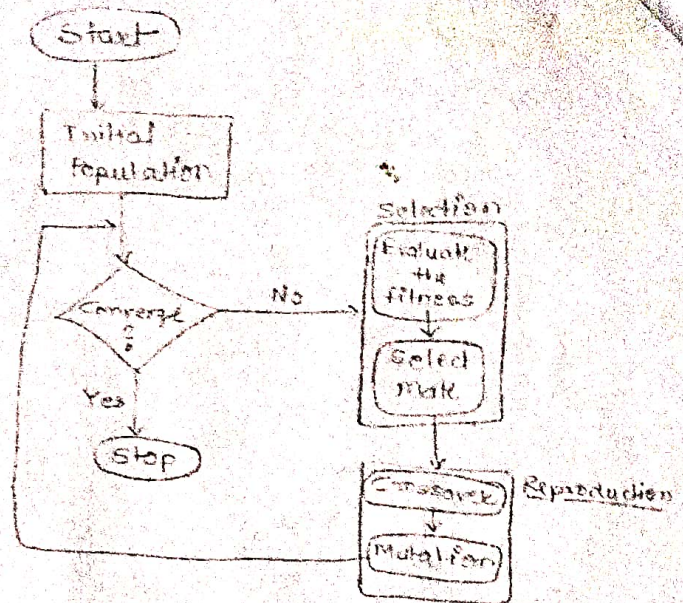
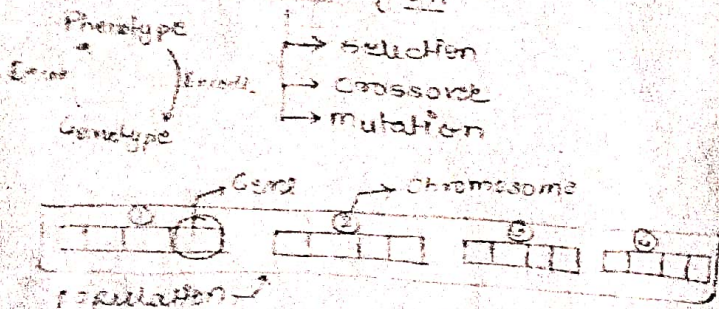
- ↳ FFN
- ↳ CNN
- ↳ RNN
- ↳ LSTM

Genetic Algorithm

- Adaptive heuristic search algo
- Inspired by "DCE" (Genetics & natural selection)
- To generate high quality solⁿ for optimization problem

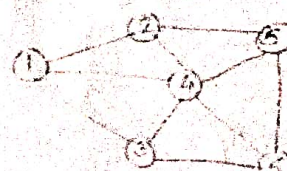
Terms

- Population
- Chromosomes
- Gene
- Allele
- Fitness function
- Operator of GA



Fitness Function

eg:-



- (P₁) : 1 2 5 6 4 3 2 1 → (18)
- (P₂) : 1 2 5 4 6 3 1 → (20)
- (P₃) : 1 2 4 5 6 3 1 → (12)
- (P₄) : 1 2 5 6 3 4 1 → (17)

Crossover

- (P₁) : 0 1 1 0 1 0
- (P₂) : 1 1 0 1 0 0

single point

- C₁ : 0 1 0 1 0 0
- C₂ : 1 1 1 0 1 0

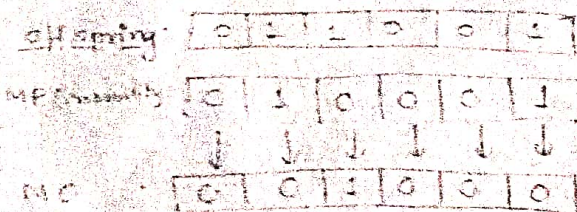
2 point

- C₁ : 0 1 0 1 1 0
- C₂ : 1 1 1 0 0 0

Multi-point

- C₁ : 1 1 0 1 1 0
- C₂ : 0 1 1 0 0 0

Mutation



or Interchanging or exchange

0 1 0 0 1 1

Skewness : measures how asymmetry of distribution of value in a dataset

Kurtosis : measures the tallness of the distribution of value in a dataset which one to use

Mild skew (-1 to 1) : No need to transformation

Moderate skew (-2 to -1 or 1 to 2) : use cube root or square root

High skew (> 2 or < -2) : use log, Box-cox or Yeo-Johnson

Negative skew : use cube root or Yeo-Johnson